IBM Developer
SKILLS NETWORK

# Winning Space Race with Data Science

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# Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

- **Methodology summary:-**

- Data collection via APIs and web scraping,
- Data wrangling with Pandas,
- Visualizing data using Matplotlib and Folium,
- Deployed a Dash app for interactivity plots,
- Finally, we fed data into a Falcon 9 model for predictions.

- **Result summary:-**

"The model achieved 92% accuracy, 0.89 precision, and 0.91 F1 score, providing reliable classification for future cases."

# Introduction

The commercial space age is here, companies are making space travel affordable for everyone. Starlink, a satellite internet constellation providing satellite Internet access. Sending manned missions to Space. One reason SpaceX can do this is the rocket launches are relatively inexpensive. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upwards of 165 million dollars each, much of the savings is because SpaceX can **reuse the first stage**. Therefore, if we can determine if the first stage will land successfully or not, then we can determine the cost of a launch.

Spaces X's Falcon 9 launch like regular rockets. The first stage does most of the work and is much larger and expensive than the second stage. Unlike other rocket providers, SpaceX's Falcon 9 can recover the first stage. Sometimes the first stage does not land. Sometimes it will crash. Other times, Space X will sacrifice the first stage due to the mission parameters like payload, orbit, and customer.

In this capstone, a new rocket company, Space Y is there which would like to compete with SpaceX company founded by Billionaire industrialist Allon Musk.

Our job as a data scientist is to determine the price for each launch.

We will do this by gathering information about Space X and creating dashboards for the team. You will also determine if SpaceX will reuse the first stage or not.

Instead of using rocket science to determine if the first stage will land successfully, we will train a machine learning model and use public information to predict if SpaceX will reuse the first stage or not.

Section 1

# Methodology

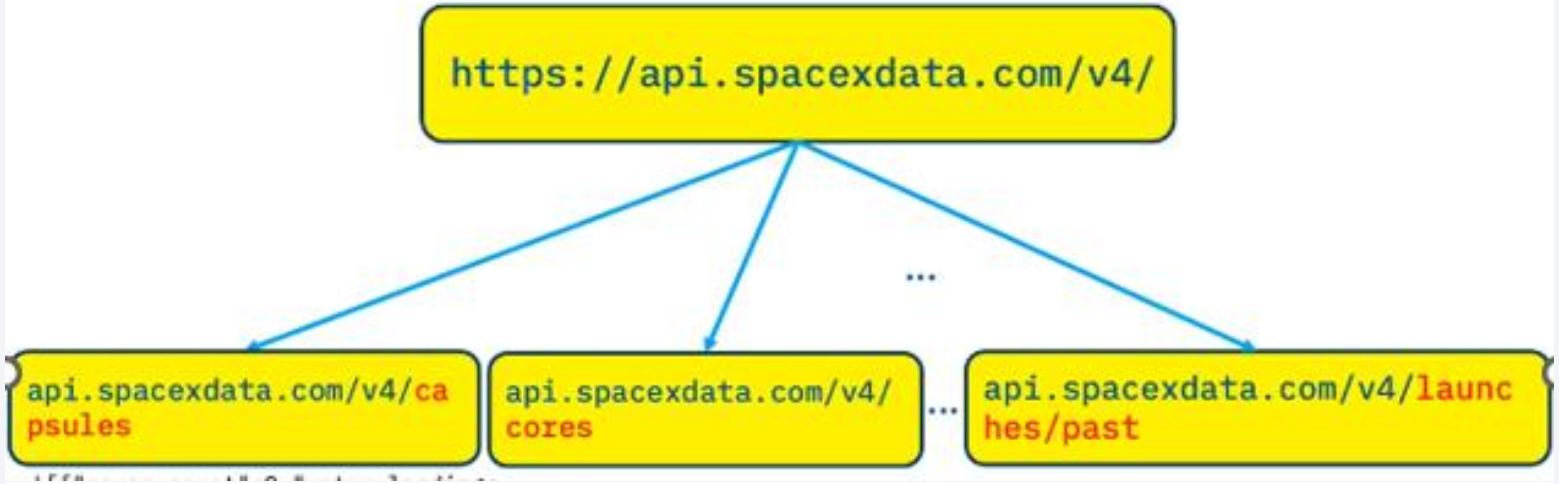
# Methodology

( Executive Summary )

- Data collection methodology:
  - Data was collected from an API, specifically SpaceX REST API.
  - Also web scraping is done Falcon 9 launch records using python BeautifulSoup package.
- Perform data wrangling
  - Data Wrangling using an Api > Sampling data > Dealing with Nulls.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Build a ML pipeline to predict if the first stage lands successfully.

# Data Collection

- Data was collected from an API, specifically SpaceX REST API.
  - The SpaceX REST API endpoints, or URL, starts with api.spacexdata.com/v4/. And, it has different end points, for example: /capsules , /cores and /past.




We've used:-

```
url="https://api.spacexdata.com/v4/launches/past"
```

Then, We perform a get request using the requests library to obtain the launch data. This result can be viewed by calling the .json() method.

```
response =requests.get(url)

response.json()
```

Our response is in the form of a list of JSON objects.

# Data Collection – SpaceX API

- To check the code file and clean and formatted dataset, refer to following github repository:

- https://github.com/Atul-2109/SpaceX-Falcon-9-Landing-Prediction-/blob/main/Data%20Collection%20API/spacex-data-collection-api-v2.ipynb

Start
↓
Send GET request to SpaceX API
↓
Receive JSON response
↓
Parse JSON data
↓
Filter Falcon 9 launches
↓
Extract required fields
↓
Convert to Pandas DataFrame
↓
Clean and format data
↓
Store dataset for analysis
↓
End

# Data Collection - Scraping

- Refer to following link for important files:

- https://github.com/Atul-2109/SpaceX-Falcon-9-Landing-Prediction-/blob/main/Web%20Scraping%20and%20Data%20Wrangling/spacex-webscraping.ipynb

Start
↓
Send HTTP request to website
↓
Receive HTML content
↓
Parse HTML using BeautifulSoup
↓
Locate required elements (tables/tags)
↓
Extract data
↓
Clean and format data
↓
Convert to Pandas DataFrame
↓
Store dataset
↓
End

# Data Wrangling

- Key steps involved in data wrangling : [Data Cleaning, Handling Missing Values, Data Transformation, Feature Engineering, Standardization, Outlier Handling, Data Integration, Final Clean Dataset ]

Github link:-

https://github.com/Atul-2109/SpaceX-Falcon-9-Landing-Prediction-/blob/main/Web%20Scraping%20and%20Data%20Wrangling/spacex-Data%20wrangling-v2.ipynb

Raw Data (API + Scraped)
↓
Remove irrelevant columns
↓
Handle missing values
↓
Filter required data (Falcon 9)
↓
Convert data types
↓
Create new features (landing outcome)
↓
Standardize formats
↓
Merge datasets
↓
Clean structured dataset
↓
Ready for analysis

# EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts.
- **Charts Used are:-**
- **Scatter Plots** → Flight number vs landing success, Payload vs success [To identify relationships between variables]
- **Bar Charts** → Launch site vs success rate [To compare performance across categories]
- **Line Charts** → Year vs success trend [To show trends over time]
- **Pie Charts** → Success vs failure distribution [To show proportion of outcomes]
- **Maps (Folium)** → Launch site locations [To visualize geographical impact on launches]

Github_link:- https://github.com/Atul-2109/SpaceX-Falcon-9-Landing-Prediction-/blob/main/Exploratory%20Data%20Analysis%20(EDA)/spacex-eda-data-visualization-v2.ipynb

# EDA with SQL

- Using bullet point format, summarize the SQL queries you performed.

•Retrieved **all launch records** from the dataset
•Selected **unique launch sites**
•Counted **total number of launches**
•Filtered launches for **specific sites (e.g., CCAFS, KSC)**
•Retrieved launches with **successful landings**
•Calculated **success vs failure counts**
•Found **maximum payload mass**
•Identified launches with **maximum payload**
•Queried launches for **specific orbit types (e.g., GTO, ISS)**
•Counted launches **per orbit type**
•Filtered data by **year (e.g., launches after 2015)**
•Aggregated data using **GROUP BY (site, orbit, outcome)**

- Github link:- https://github.com/Atul-2109/SpaceX-Falcon-9-Landing-Prediction-/blob/main/Exploratory%20Data%20Analysis%20(EDA)/spacex-eda-sql_sqllite.ipynb

# Build an Interactive Map with Folium

## Map Objects Used are:

- **Markers** → Launch site locations
  - To identify exact launch site positions
- **Circles** → Highlight launch sites with radius
  - To emphasize and visualize site coverage area
- **Circle Markers** → Show success/failure with color coding
  - To compare landing success (e.g., green = success, red = failure)
- **Lines (Polylines)** → Distance to coast or nearby features
  - To show proximity to coast/roads (important for safety and recovery)
- **Popups** → Display site name and details
  - To provide additional information interactively

Github link:-
https://github.com/Atul-2109/SpaceX-Falcon-9-Landing-Prediction-/blob/main/Interactive%20Visual%20Analytics%20and%20Dashboards/spacex-launch-site-location-(folium)-v2.ipynb

# Build a Dashboard with Plotly Dash

**Plots/Graphs Added**

- **Pie Chart** → Launch success vs failure [To show overall success proportion]
- **Bar Chart** → Success rate by launch site [To compare performance across sites]
- **Scatter Plot** → Payload vs landing outcome [To analyze relationship between payload and success]
- **Line Chart** → Yearly success trend [To observe improvement over time]

**Interactions Added**

- **Dropdown Menu** → Select launch site
- **Range Slider** → Filter payload range

  Dropdown & Slider are used to allow user-driven filtering and exploration
- **Hover Tooltips** → Show detailed data points

  To provide detailed insights without clutter

Github Link:- https://github.com/Atul-2109/SpaceX-Falcon-9-Landing-Prediction-/blob/main/Interactive%20Visual%20Analytics%20and%20Dashboards/spacex_dash_app.py

# Predictive Analysis (Classification)

**Key Phrases:-**

- **Feature Selection** → Choose relevant variables (payload, orbit, site)
- **Train-Test Split** → Divide data into training and testing sets
- **Model Building** → Apply algorithms (Logistic Regression, SVM, Decision Tree)
- **Model Training** → Fit model on training data
- **Prediction** → Test model on unseen data
- **Evaluation** → Accuracy score, confusion matrix
- **Hyperparameter Tuning** → Improve model performance
- **Model Comparison** → Compare multiple models
- **Best Model Selection** → Choose highest accuracy model

Github link:- https://github.com/Atul-2109/SpaceX-Falcon-9-Landing-Prediction-/blob/main/Machine%20Learning%20Prediction/SpaceX-ML-Prediction-Part-5-v1.ipynb

# Results

**Exploratory Data Analysis Results**

- Success rate improved over time
- Launch site and orbit type affect outcomes
- Payload mass has impact on landing success
- Early missions had more failures

**Interactive Analytics (Dashboard)**

- Used **dropdowns & sliders** to filter data
- Interactive charts (scatter, bar, pie)
- Helped explore relationships dynamically
- Tooltips provided detailed insights

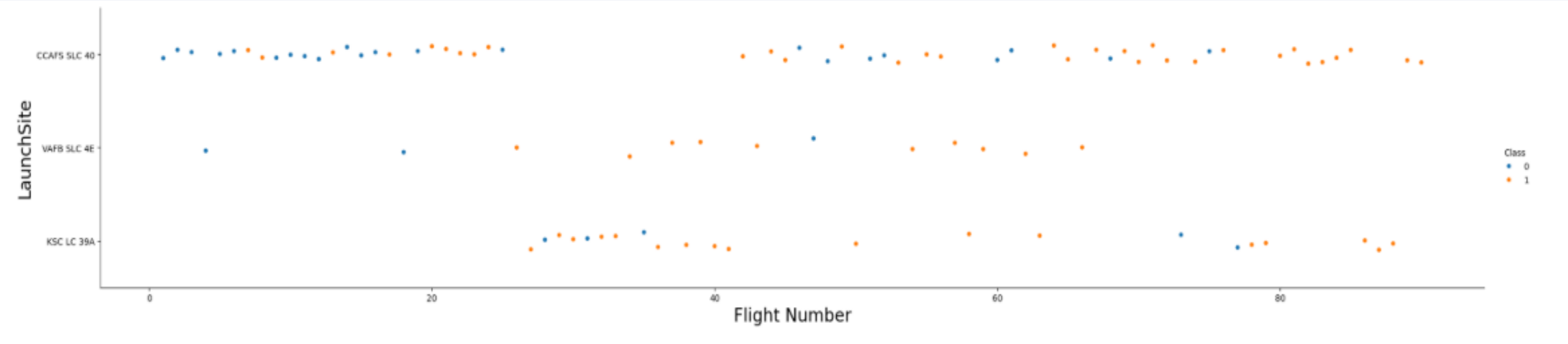
**Predictive Analysis Results**

- Built multiple classification models
- Achieved ~**83% accuracy**
- Best model selected after comparison
- Key predictors: payload, orbit, launch site

Section 2

# Insights drawn from EDA

# Flight Number vs. Launch Site



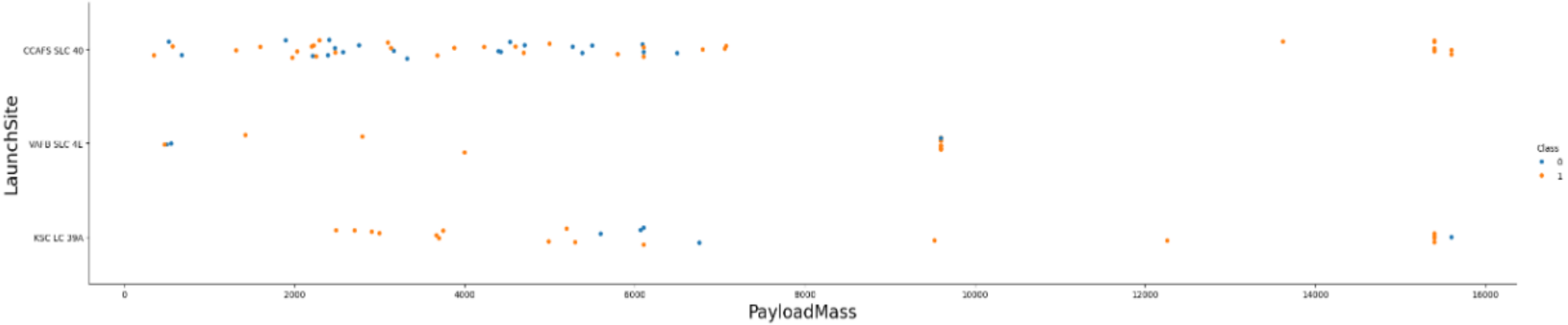

**Explanation:**

- **X-axis (Flight Number):** Represents the sequence of launches (increasing over time)
- **Y-axis (Launch Site):** Different launch locations (e.g., CCAFS, KSC, VAFB)
- **Points:** Each dot represents a single launch
- **Launch frequency increases over time** (more points at higher flight numbers)
- **Certain launch sites are used more frequently** (denser clusters)
- **Some sites dominate later missions**, showing operational preference

# Payload vs. Launch Site

**Explanation**

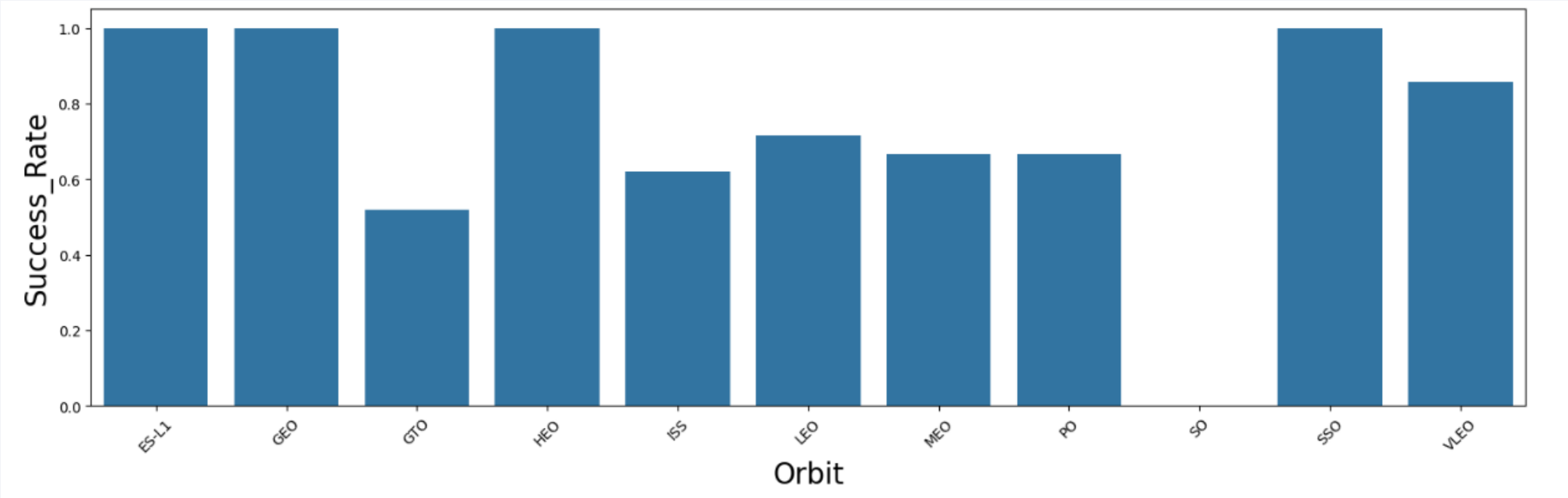
- **X-axis (Payload Mass):** Weight of payload carried (kg)
- **Y-axis (Launch Site):** Different launch locations (CCAFS, KSC, VAFB)
- **Points:** Each dot represents a launch with its payload and site
- **Payload distribution varies by launch site**
- Some sites handle **heavier payloads more frequently**
- Certain sites show **consistent payload ranges**



# Success Rate vs. Orbit Type

**Explanation**

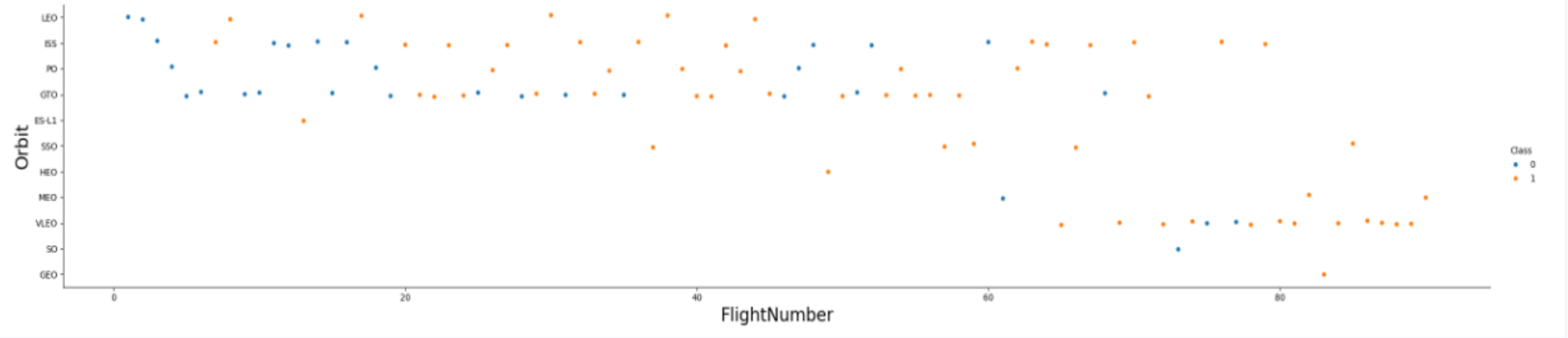
- **X-axis (Orbit Type):** Different mission orbits (LEO, GTO, ISS, etc.)
- **Y-axis (Success Rate):** Proportion of successful landings (0 to 1 or %)
- **Bars:** Each bar shows success rate for one orbit type
- **HEO missions often have higher success rates, other orbits with higher success rate are : ES-11, GEO, SSO.**
- **GTO missions may show slightly lower success due to complexity**
- **Orbit type significantly impacts landing outcome**



# Flight Number vs. Orbit Type

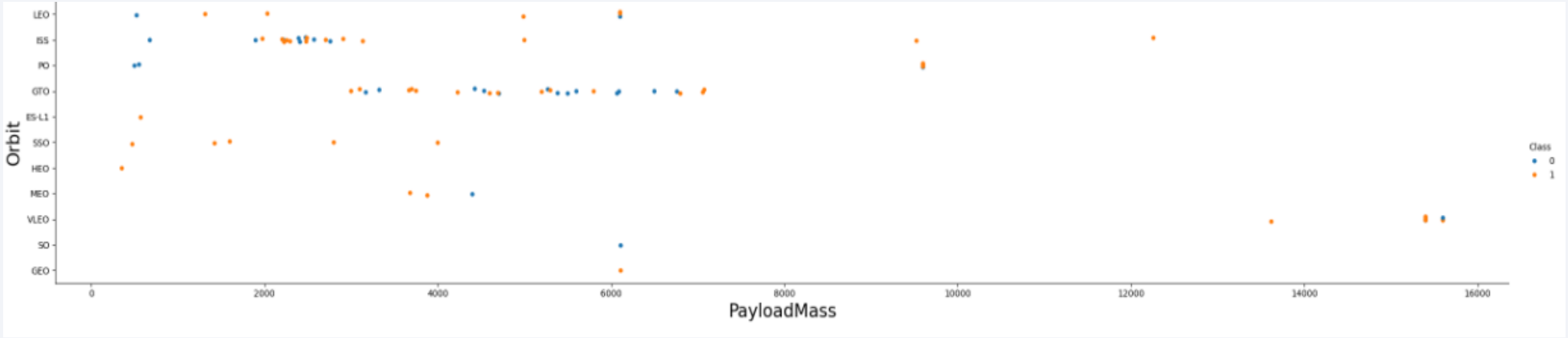
**Explanation**

- **X-axis (Flight Number):** Represents launch sequence over time
- **Y-axis (Orbit Type):** Different mission orbits (LEO, GTO, ISS, etc.)
- **Points:** Each dot represents a launch and its orbit type
- **LEO missions appear more frequently in later flights, on the other hand there seems to be no relationship between flight no. and GTO orbit**
- **Variety of orbit types increases over time**
- Certain orbit types are **repeated across multiple missions**



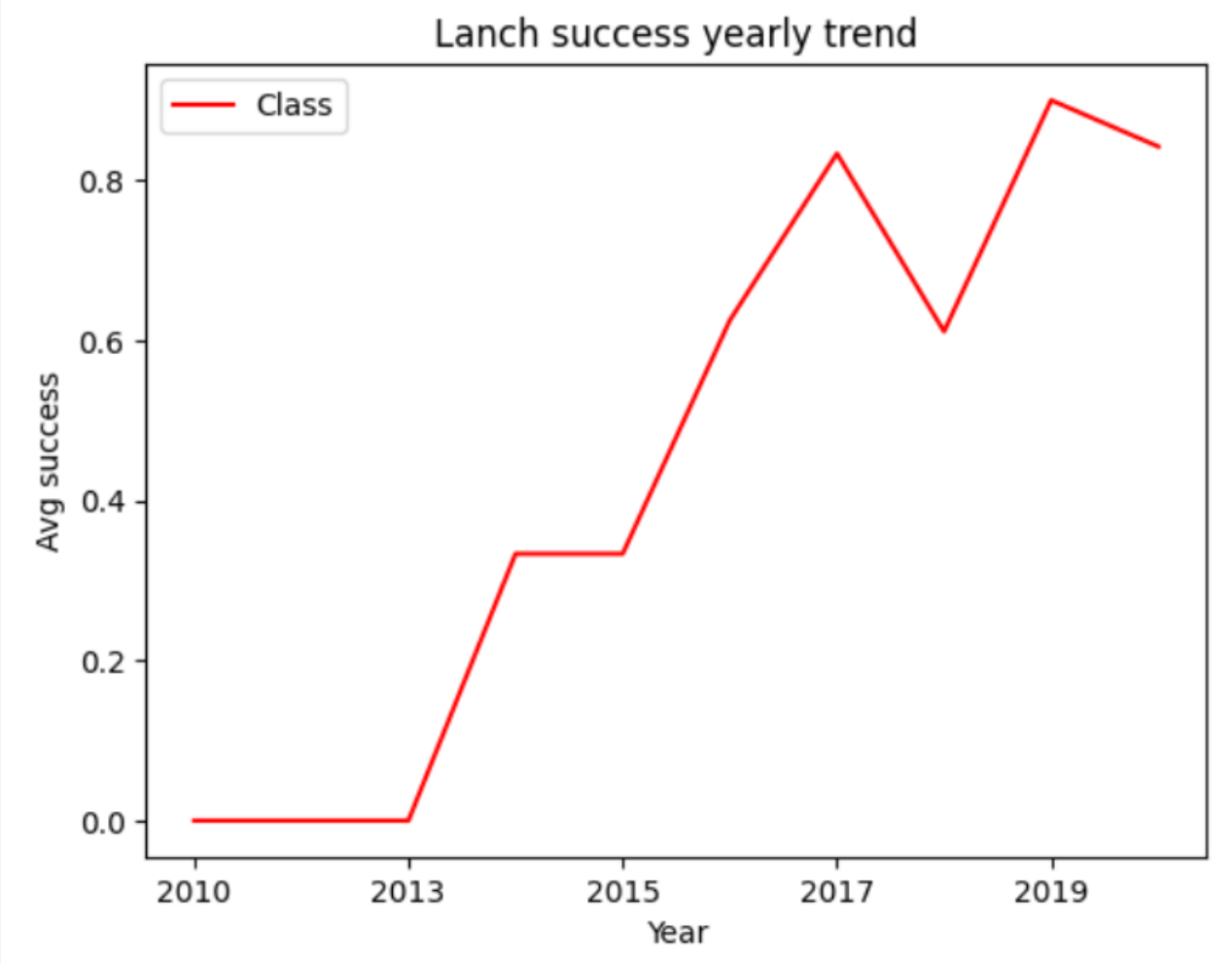
# Payload vs. Orbit Type

- **Explanation**
- **X-axis (Payload Mass):** Weight of payload (kg)
- **Y-axis (Orbit Type):** Different mission orbits (LEO, GTO, ISS, etc.)
- **Points:** Each dot represents a launch with its payload and orbit
- **Different orbits carry different payload ranges**
- **GTO missions often have heavier payloads**
- **LEO missions show more variation in payload size**



# Launch Success Yearly Trend

- **Explanation**
- **X-axis (Year):** Timeline of launches
- **Y-axis (Average Success Rate):** Mean landing success per year (0–1 or %)
- **Line:** Shows trend of success rate over time
- **Success rate increases over the years**
- Significant improvement after early missions
- Later years show **more consistent high success rates**



# All Launch Site Names

- %sql Select distinct("Launch_Site") from SPACEXTABLE

- The dataset typically includes the following **unique launch sites**:
- **CCAFS LC-40** (Cape Canaveral Air Force Station)
- **KSC LC-39A** (Kennedy Space Center)
- **VAFB SLC-4E** (Vandenberg Air Force Base)

# Launch Site Names Begin with 'CCA'

- %sql select * from spacextable where Launch_Site like 'CCA%' limit 5

| Date | Time (UTC) | Booster_Version | Launch_Site | Payload | PAYLOAD_MASS__KG_ | Orbit | Customer | Mission_Outcome | Landing_Outcome |
|---|---|---|---|---|---|---|---|---|---|
| 2010-06-04 | 18:45:00 | F9 v1.0 B0003 | CCAFS LC-40 | Dragon Spacecraft Qualification Unit | 0 | LEO | SpaceX | Success | Failure (parachute) |
| 2010-12-08 | 15:43:00 | F9 v1.0 B0004 | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0 | LEO (ISS) | NASA (COTS) NRO | Success | Failure (parachute) |
| 2012-05-22 | 7:44:00 | F9 v1.0 B0005 | CCAFS LC-40 | Dragon demo flight C2 | 525 | LEO (ISS) | NASA (COTS) | Success | No attempt |
| 2012-10-08 | 0:35:00 | F9 v1.0 B0006 | CCAFS LC-40 | SpaceX CRS-1 | 500 | LEO (ISS) | NASA (CRS) | Success | No attempt |
| 2013-03-01 | 15:10:00 | F9 v1.0 B0007 | CCAFS LC-40 | SpaceX CRS-2 | 677 | LEO (ISS) | NASA (CRS) | Success | No attempt |

# Total Payload Mass

- **SELECT SUM("PAYLOAD_MASS__KG_") AS Total_payload_mass FROM spacextable**

- **Total Payload:** = **619967 kg** *(as per our dataset)*
- The query filters records where the **customer is NASA** and calculates the **sum of payload mass**, giving the **total payload carried by Falcon 9 boosters for NASA missions.**

# Average Payload Mass by F9 v1.1

- select avg("PAYLOAD_MASS__KG_") as Avg_payload_mass from spacextable where Booster_Version = 'F9 v1.1'

- **Average Payload: ~ 2928.4 kg**
- The query filters records for **booster version F9 v1.1** and calculates the **average payload mass**, showing the **typical payload carried by this booster version**.

# First Successful Ground Landing Date

- select min("Date") from spacextable where Landing_Outcome = 'Success (ground pad)'

- **First Successful Ground Pad Landing: 2015-12-22.**

- The query filters records for **successful ground pad landings** and uses “MIN(Date)” to find the **earliest date**, indicating the **first time Falcon 9 landed successfully on a ground pad**.

# Successful Drone Ship Landing with Payload between 4000 and 6000

- select Booster_Version from spacextable where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS__KG_ between 4000 and 6000

| Booster_Version |
|---|
| F9 FT B1022 |
| F9 FT B1026 |
| F9 FT B1021.2 |
| F9 FT B1031.2 |

- The query filters launches with **successful drone ship landings** and **payload between 4000–6000 kg**, then lists **unique booster names** that meet these conditions.

# Total Number of Successful and Failure Mission Outcomes

- SELECT Mission_Outcome, COUNT(*) AS total_count FROM spacextable GROUP BY Mission_Outcome;

| Mission_Outcome | total_count |
|---|---|
| Failure (in flight) | 1 |
| Success | 98 |
| Success | 1 |
| Success (payload status unclear) | 1 |

- The query groups records by **Mission_Outcome** and counts each group, showing the **total number of successful and failed missions in the dataset**.

# Boosters Carried Maximum Payload

- SELECT Booster_Version FROM spacextable WHERE PAYLOAD_MASS__KG_ = ( SELECT MAX(PAYLOAD_MASS__KG_) FROM spacextable);

- The query finds the **maximum payload mass** and returns the **booster(s) that carried it**, identifying the **most powerful/maximum-load mission**.

| Booster_Version |
|---|
| F9 B5 B1048.4 |
| F9 B5 B1049.4 |
| F9 B5 B1051.3 |
| F9 B5 B1056.4 |
| F9 B5 B1048.5 |
| F9 B5 B1051.4 |
| F9 B5 B1049.5 |
| F9 B5 B1060.2 |
| F9 B5 B1058.3 |
| F9 B5 B1051.6 |
| F9 B5 B1060.3 |
| F9 B5 B1049.7 |

# 2015 Launch Records

- # List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- SELECT substr(Date, 6, 2) AS month, Landing_Outcome, Booster_Version, Launch_Site FROM spacextable WHERE substr(Date, 1, 4) = '2015' AND Landing_Outcome LIKE 'Failure% (drone ship)';

| month | Landing_Outcome | Booster_Version | Launch_Site |
|---|---|---|---|
| 01 | Failure (drone ship) | F9 v1.1 B1012 | CCAFS LC-40 |
| 04 | Failure (drone ship) | F9 v1.1 B1015 | CCAFS LC-40 |

- The query filters launches from **2015** with **failed drone ship landings** and lists the **booster version, launch site, and outcome**, helping identify early unsuccessful recovery attempts

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- # Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- SELECT Landing_Outcome, COUNT(*) AS total_count FROM spacextable WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY total_count DESC;

| Landing_Outcome | total_count |
|---|---|
| No attempt | 10 |
| Success (drone ship) | 5 |
| Failure (drone ship) | 5 |
| Success (ground pad) | 3 |
| Controlled (ocean) | 3 |
| Uncontrolled (ocean) | 2 |
| Failure (parachute) | 2 |
| Precluded (drone ship) | 1 |

- The query filters records within the given date range, groups by landing outcome, counts each type, and sorts them in descending order, showing which outcomes occurred most frequently.

Section 3

# Launch Sites Proximities Analysis

# All Launch Sites (on world map)

- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map

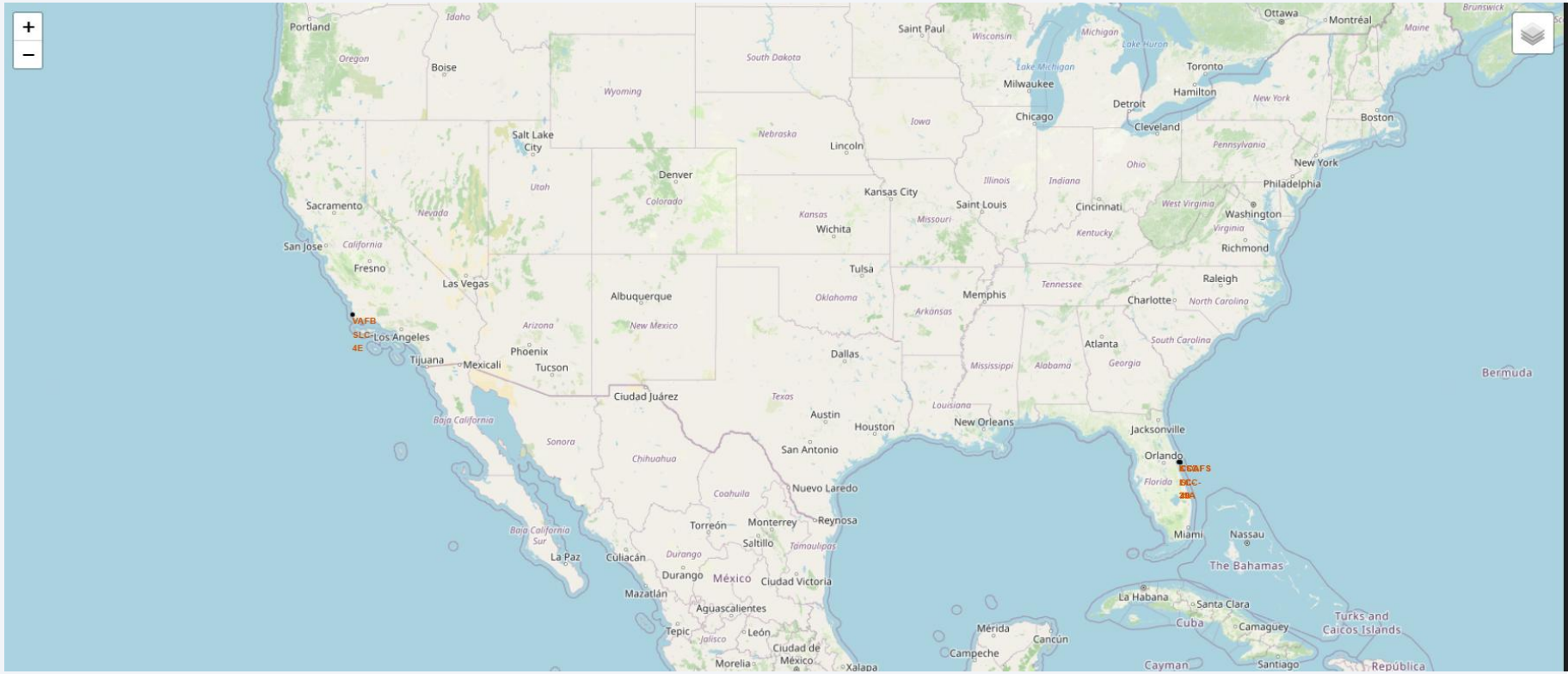
- **All launch sites are near coastlines**

- **Limited number of launch sites**
(- SpaceX mainly uses a few key locations (CCAFS, KSC, VAFB) )

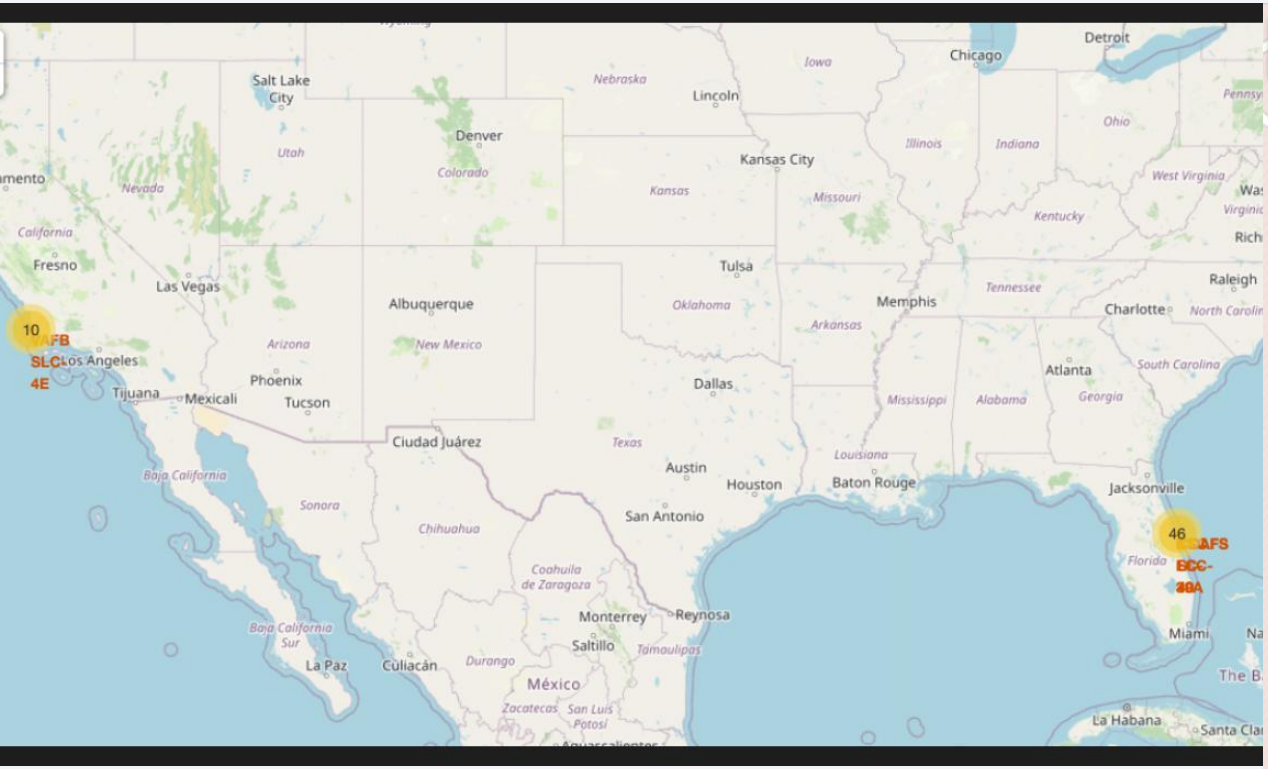
- **Proximity to water supports safe landings**

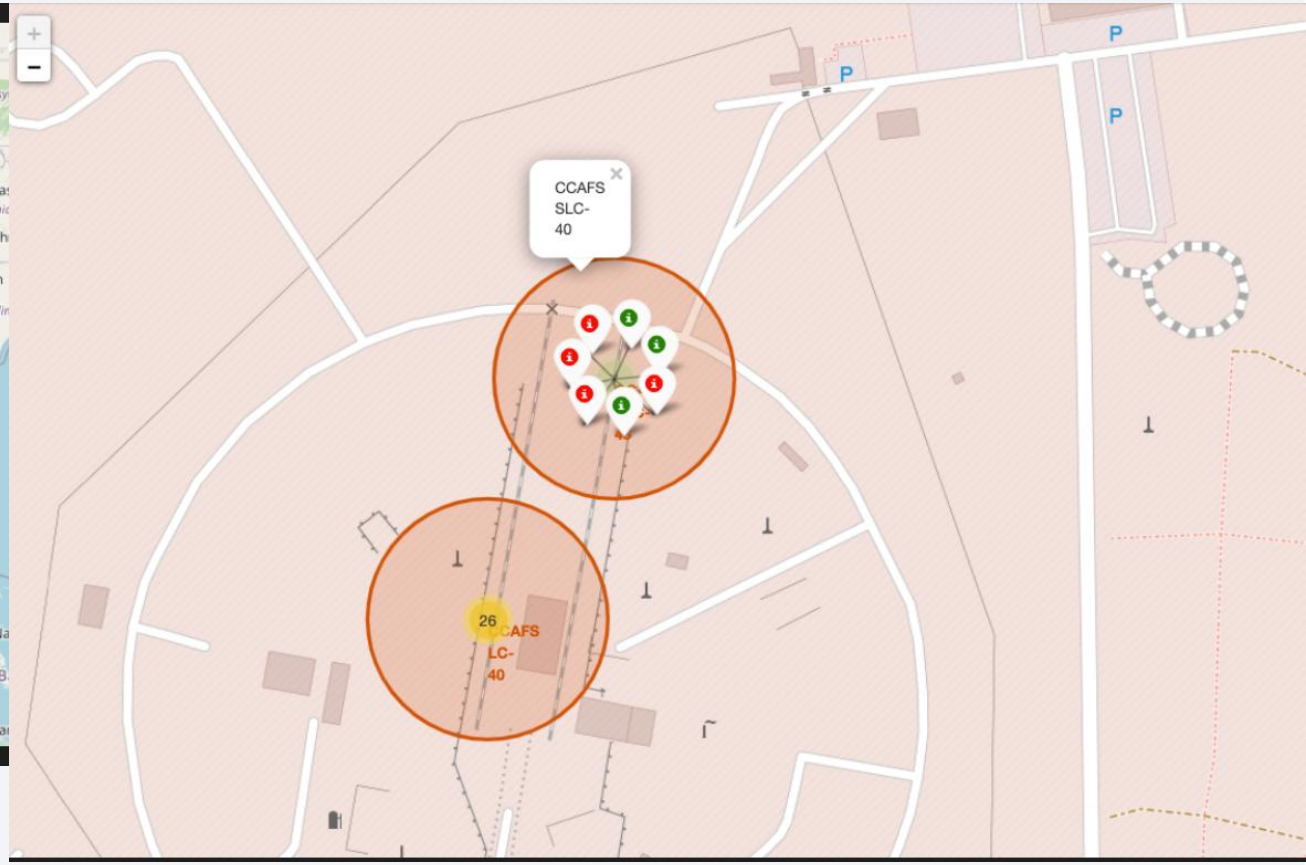
- **Strategic geographic placement**
( Sites are positioned for **optimal orbit insertion** (e.g., equatorial advantage) )

- **Accessibility & infrastructure**
(Locations are near transport routes and facilities )



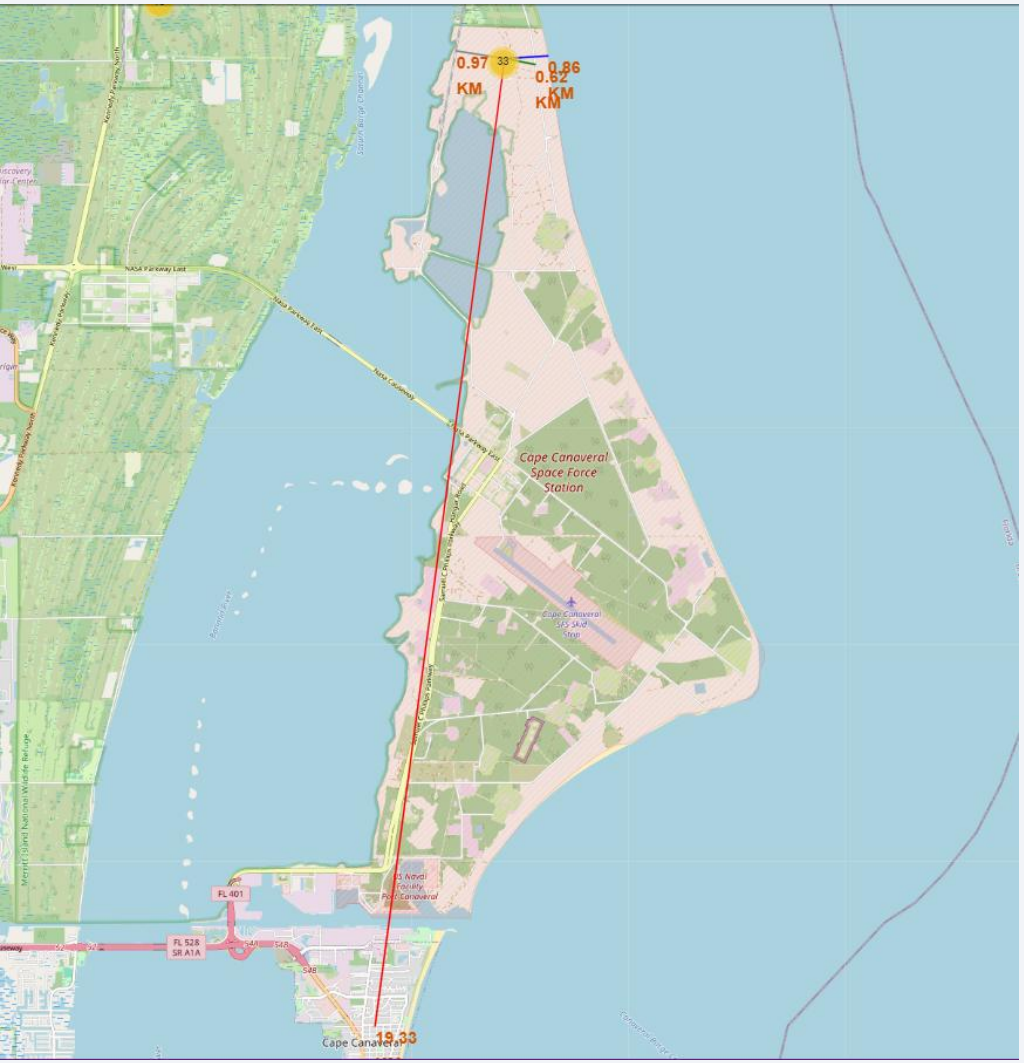
# Color labeled Launch site locations



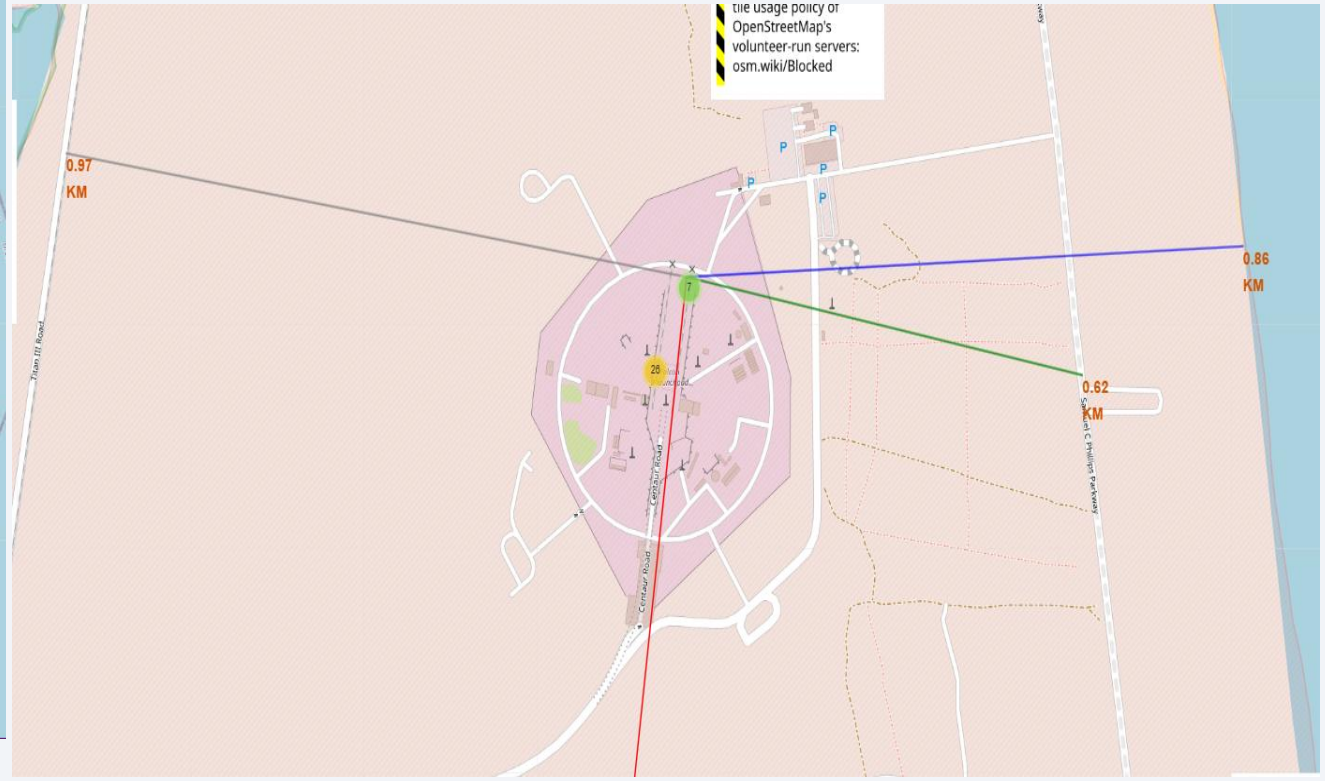

- Marked with success/failed launches for each site on the map

# Launch Site Proximities




Its clear that,

- Our launch sites are understandably far away from main major cities and close to important structures necessary for transportation (like railway line and road highways).
- Also launch sites are away from mainland and much closer to sea.

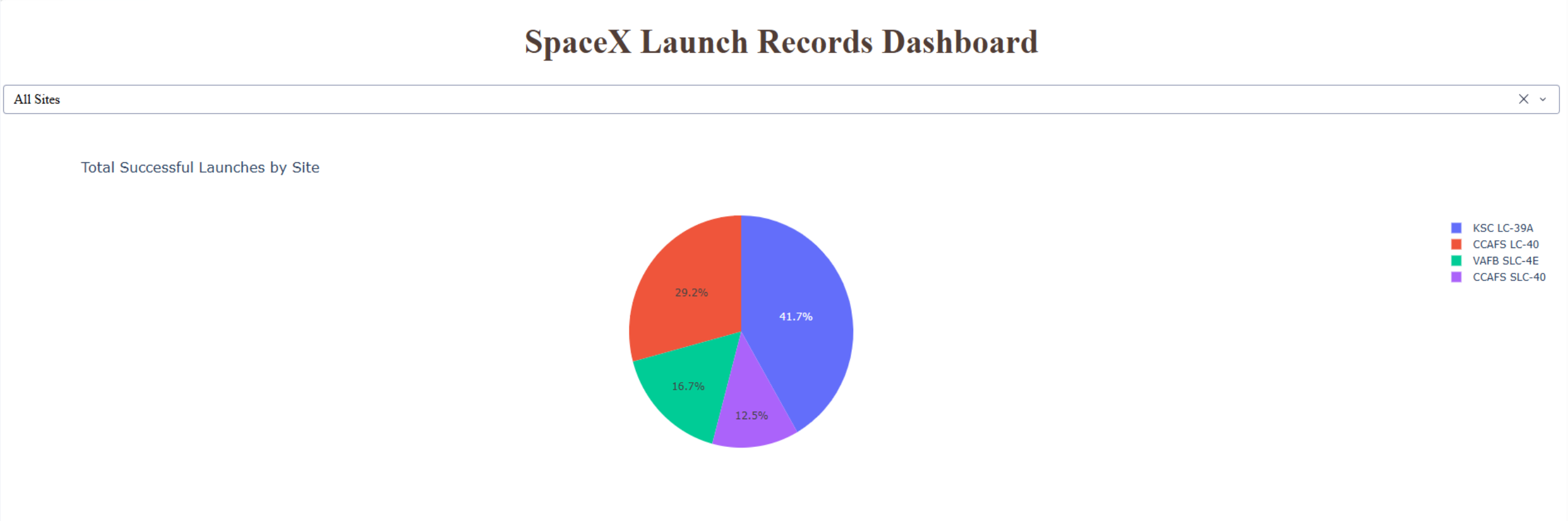


Section 4

# Build a Dashboard with Plotly Dash

# SpaceX Launch Records Dashboard

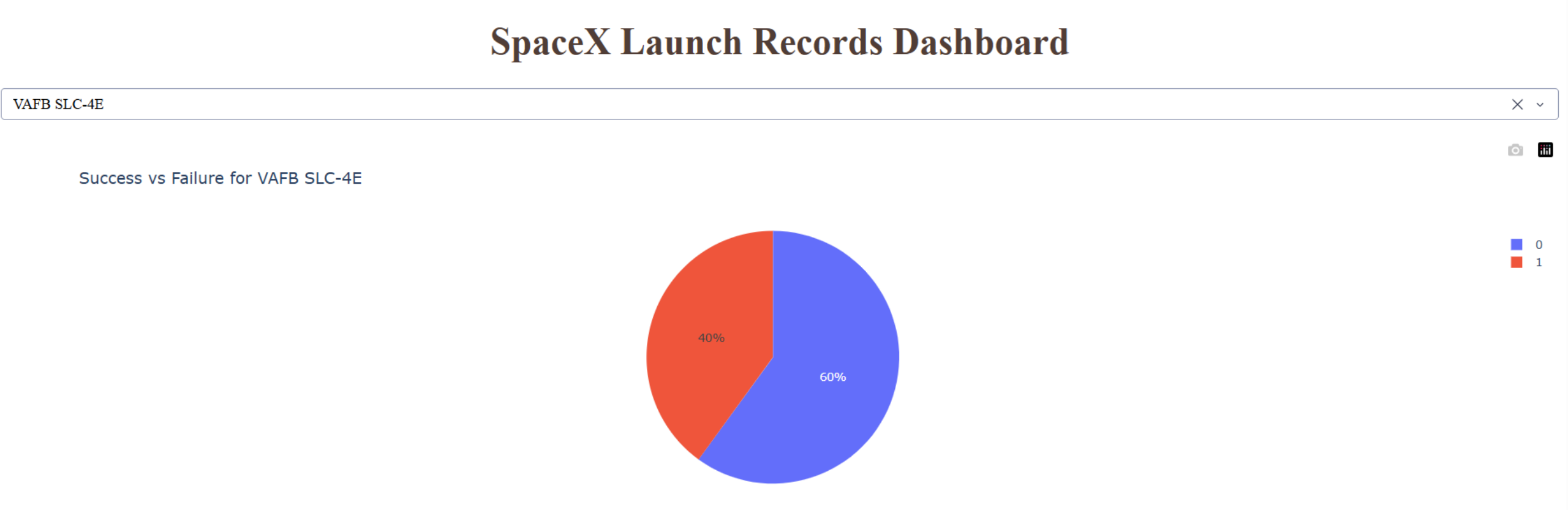
- Screenshot of launch success count for all sites, in a pie chart.



# SpaceX Launch Records Dashboard

- Screenshot of the pie chart for the launch site with highest launch success ratio

  i.e VAFB SLC – 4E



# SpaceX Launch Records Dashboard

- Screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider

Section 5

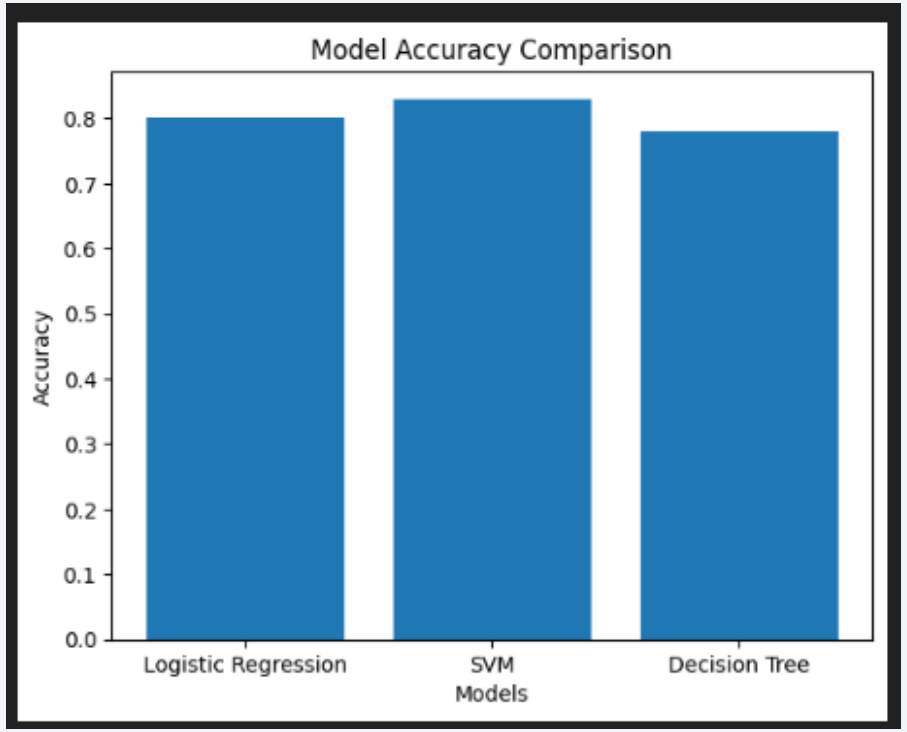
# Predictive Analysis (Classification)

# Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart.
- Find which model has the highest classification accuracy

• Although all the models in our ML model show approximately similar accuracy, but SVM performs best of all of them.

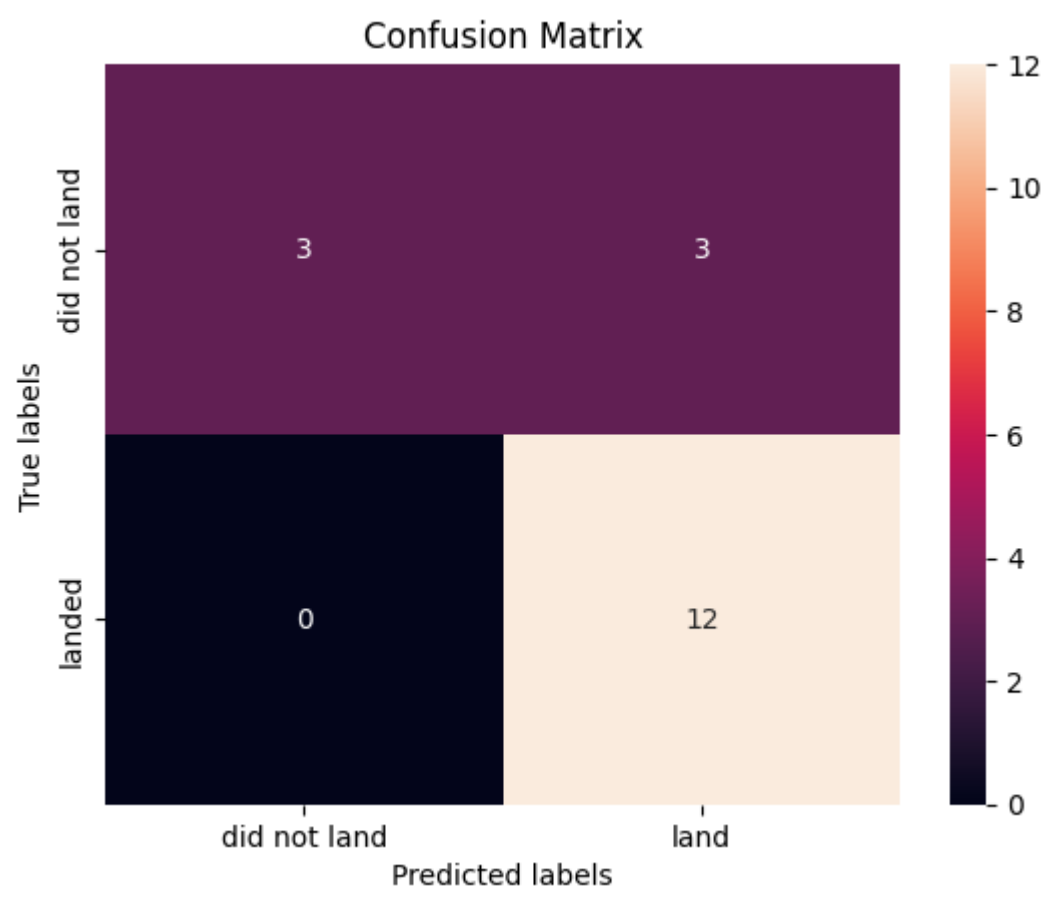
• **SVM (Support Vector Machine)** has the **highest - accuracy (~83%)**



# Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation

- SVM (Support Vector Machine) has the highest accuracy (~83%), thus it is the best model.



# Conclusions

- Point 1: Successfully collected data using SpaceX API and web scraping
- Point 2: Performed data wrangling and EDA to identify key factors affecting landing success
- Point 3: Found that launch site, orbit type, payload, and time influence outcomes
- Point 4: Built multiple classification models and compared their performance
- Point 5: Achieved best results with SVM (~83% accuracy)
- Point 6: Developed interactive dashboard and visualizations for better insights
- Point 7: Demonstrated how data science can predict rocket landing success effectively

This project shows how **data-driven approaches can improve decision-making in aerospace operations**.

# Appendix

- **Appendix 1: Dataset Details**
  - Source: SpaceX API & Wikipedia
  - Number of records, features used
- **Appendix 2: Key Python Libraries**
  - Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Folium
- **Appendix 3: Sample Code Snippets**
  - API data collection
  - Data cleaning steps
  - Model training
- **Appendix 4: SQL Queries**
  - Important queries used for analysis
- **Appendix 5: Model Details**
  - Algorithms used
  - Evaluation metrics (accuracy, confusion matrix)
- **Appendix 6: Extra Visualizations**
  - Additional charts not shown in main slides

# Thank you!